Jimmy Dong

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EDUCATION

B.S., Double Major in Computer Engineering Tech. and Electrical Engineering Tech. **May '28**

Rochester Institute of Technology | Rochester, NY

Coursework: **Electronic Devices** (In Progress), **Microcontroller System** (In Progress), **Digital Systems Design**, **Circuits 1**, **Circuits II** , **Computational Problem Solving**, **Fundamentals of Engineering**, **Intro to Digital & Micro Systems**

SKILLS

Technical Skills: FPGA (DEO-CV Board), MSP430 Microcontroller, Embedded Systems, Digital Logic Design, Circuit Analysis, Signal Processing, Prototyping, Soldering

Computer Skills: VHDL, C/C++, Python, Java, R, MATLAB, Microsoft Office 365, Altium, Quartus, ModelSim

WORK EXPERIENCE

Research Intern – Environmental Data Systems **Jun '24 - Aug '24**

UNC Chapel Hill | Chapel Hill, NC

- Optimized research on **environmental monitoring technologies** by integrating **sensors** and **data-acquisition hardware**, resulting in a **15% enhancement in data accuracy**
- Delivered in-depth analysis of **real-time environmental data** with **laboratory measurement systems**, enhancing predictive accuracy and **improving resource allocation efficiency by 20%**
- Collaborated with **Morehead Planetarium** to explore applications of **sensor-based monitoring** and **environmental data visualization**

PROJECTS

Electric Vehicle Team (EVT) - Wire Harness Team **Sep '25 - Present**

- Integrated and routed **wire harness systems** to reduce **signal interference** and improve **electrical reliability**
- Performed **continuity testing, voltage verification,** and **signal integrity validation** to ensure compliance with **SAE safety and performance standards**
- Assisted in diagnosing and resolving **electrical faults**, including **connector failures** and **signal inconsistencies** within **drivetrain** and **control modules**

Vending Machine Controller (Finite State Machine Design) **Oct '25 - Dec '25**

- Designed a **finite state machine**–based vending machine controller that accepts **nickels, dimes, and quarters** to dispense a **$0.75 product** with correct **change handling**
- Refined **transaction control logic** using **VHDL**, enhancing **dispense validation**, **coin return functionality**, and **deposit rejection processes**, which elevated operational efficiency by **10%**
- Displayed **real-time balance** using **dual seven-segment displays**, providing continuous **user feedback** throughout the transaction

Parallel Resonance Band-Pass Filter Design **Sep '25 - Oct '25**

- Designed and analyzed a **parallel resonant band-pass filter** using **ideal and non-ideal inductor models** to determine **resonant frequency**, **Q factor**, **bandwidth**, and **voltage gain**
- Performed **AC frequency sweep simulation** to identify **resonance** and **3 dB cutoff frequencies**, comparing **theoretical** and **simulated results**
- Built and tested the physical circuit, collecting **50+ frequency response measurements** to evaluate the impact of **inductor series resistance** on **Q factor** and **bandwidth**

7-Segment Display Controller (MSP430 Microcontroller) **Feb '25 - Apr '25**

- Programmed an **MSP430 microcontroller in C** to drive a **multiplexed 4-digit seven-segment display via GPIO and timer-based refresh logic**
- Executed **digit extraction**, **timing control**, and **display refreshing algorithms** for **real-time numeric output** with **zero visible flicker**
- Enhanced **display control responsiveness** by **20%** through **switch integration** and **multi-scenario timing validation**